

SkyFusion: A Physics-Informed GAN-Transformer Framework for Intra-Hour Sky Forecasting and PV Prediction

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Abstract: Reliable integration of renewable energy sources into electrical grids hinges on accurate forecasting of sky conditions and photovoltaic (PV) power generation [Nie23]. However, existing methods, whether traditional numerical weather prediction or current deep learning models, have faced challenges in reliably generating realistic intra-hour sky conditions (e.g., at 15-minute horizons) and associated PV power forecasts, primarily due to the complex and stochastic dynamics of cloud motion [Nie23]. In this work, we address this gap, demonstrating successful probabilistic sky forecasting and associated PV predictions at a 15-minute horizon. To address prior limitations, we introduce **SkyFusion**, a novel, modular pipeline that synergistically combines physics-informed latent-space modeling (PhyCell) [LG20], transformer-based autoregression, and a multi-stage generative adversarial network (GAN) approach. Although GAN components faced initial stability challenges, clear strategies for future integration are outlined, paving the way towards high-fidelity, physically plausible intra-hour sky sequences and robust probabilistic PV power prediction.

Keywords: Solar Forecasting; PV Power; Cloud Motion; Physics-Informed NN; Generative Adversarial Networks; Transformer; ConvLSTM; VQ-VAE.

1 Introduction

The integration of solar energy into modern power grids requires accurate, short-term forecasting of photovoltaic (PV) output, especially at minute-level resolutions [Nie23, LG20]. Rapid cloud dynamics disrupt grid stability and challenge both energy trading and load balancing. Traditional NWP models lack temporal granularity, while purely data-driven methods offer limited physical interpretability and often yield deterministic outputs [Nie23, LG20]. SkyFusion addresses this by combining physics-informed modeling with deep generative learning to enable robust, probabilistic 15-minute forecasts. Chapter 5 details the ConvLSTM-UNet baseline, which forms the foundation of the forecasting pipeline.

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2 Objectives and Scope

This paper focuses on **Milestone 1**, which covers data preparation, sun-position masking (via PVLib), and the implementation of a ConvLSTM-UNet baseline that forecasts the next 15 minutes at 30-second intervals. GAN components—specifically PatchGAN—were successfully integrated into the baseline to enhance the visual quality of predictions. Their positive effect in reducing blurriness is demonstrated through side-by-side comparisons. A roadmap for future GAN extensions (e.g., TemporalGAN) is outlined to further improve realism and temporal coherence.

The structure of the paper has been revised for clarity and scientific rigor. Section 3 reviews related work; Section 4 outlines the SkyFusion pipeline; Section 5 details the ConvLSTM-UNet implementation and validation; Section 6 presents benchmark results; Section 7 discusses future work, including GAN stabilization and scalability; Section 8 concludes.

3 Relevant Research

Recent advances in sky image forecasting and PV power prediction have leveraged deep learning to address the challenges of spatiotemporal dynamics and uncertainty.

SkyGPT [Nie23] introduced a two-stage pipeline using a VQ-VAE and a physics-constrained transformer (Phy-Transformer) for probabilistic sky video generation, followed by PV power nowcasting. It employs L2 and moment losses but lacks adversarial or perceptual objectives, resulting in blurred textures. SkyFusion builds on this by integrating GAN-based perceptual training (planned in future stages) to enhance visual sharpness and extend forecasting beyond 15 minutes.

PhyDNet [LG20] proposed PhyCell, a physics-informed recurrent unit that disentangles motion from appearance, improving the physical consistency of video predictions. However, it lacks adversarial refinement, often yielding blurred outputs. SkyFusion adopts similar physics-based constraints while planning GAN integration to address visual fidelity.

ConvLSTM [SH15] remains a standard for spatiotemporal video prediction but suffers from blurriness and temporal degradation in long-range forecasting. SkyFusion uses a ConvLSTM-UNet baseline enhanced with multi-scale SSIM loss to address these limitations.

GANs [IS17] improve image realism in video forecasting by enforcing local (PatchGAN) and temporal (TemporalGAN) coherence. While unstable during training, their integration with structured models can significantly enhance visual quality. SkyFusion’s GAN module shows a finite resolution in new predicted pictures (see Section 6).

4 Project Overview

Accurate intra-hour PV power forecasting is vital for grid stability under high renewable penetration. **SkyFusion** addresses this by combining four modular deep learning stages that forecast sky conditions 15 minutes ahead and convert them into probabilistic PV power estimates.

4.1 Stage1 ConvLSTM-UNet Baseline Generation (SkyBuilder)

Captures short-term cloud motion and forecasts 30 frames (15 minutes at 30s intervals) using L1 and SSIM losses. Inputs are downsampled to 128×128 for efficiency. PatchGAN is applied for texture enhancement.

4.2 Stage2 Latent Tokenisation with VQ-VAE (SkyCompressor)

Compresses sky video clips into discrete $16 \times 16 \times T$ latent tokens using a shared codebook ($K \approx 1024$).

4.3 Stage3 Transformer Forecasting with Physics Constraints (SkyForecaster)

Uses a GPT-style transformer with PhyCell to forecast latent tokens autoregressively. PatchGAN and optional TemporalGAN aim to improve spatial and temporal coherence.

4.4 Stage4 PV NowCasting and Energy Forecasting (SkyNowcaster)

Maps the predicted sky frame at $t+15$ to a probabilistic PV power output using a CNN regressor trained with RMSE and CRPS losses. This supports uncertainty-aware grid decision-making.

Each module contributes toward generating physically consistent, visually realistic sky videos and linking them to operational PV forecasting.

5 Milestone 1: Data Preparation & Baseline

This section details the **first milestone** of the SkyFusion project, focused on preparing a high-quality, physics-consistent dataset and implementing a functional **ConvLSTM-UNet baseline** for intra-hour sky video forecasting. This foundational step establishes a reproducible benchmark to evaluate our data pipeline, modeling infrastructure, and early forecasting capabilities.

5.1 Dataset Cleaning and Image Selection

The first milestone of SkyFusion focuses on building a high-quality, physics-consistent dataset and implementing a ConvLSTM-UNet baseline for intra-hour sky video forecasting. Starting with over 400,000 raw images from two ASI cameras (asi_16030 and asi_16515) at a German PV test field [AI24], we applied a rigorous preprocessing pipeline. This included removal of corrupted frames and calibration artifacts, while preserving minor imperfections like droplets to retain real-world variability. The original 15-second interval was downsampled to 30 seconds to reduce redundancy and improve temporal coherence. The resulting dataset of approximately 178,000 images serves as a robust foundation for model training and reproducible benchmarking.

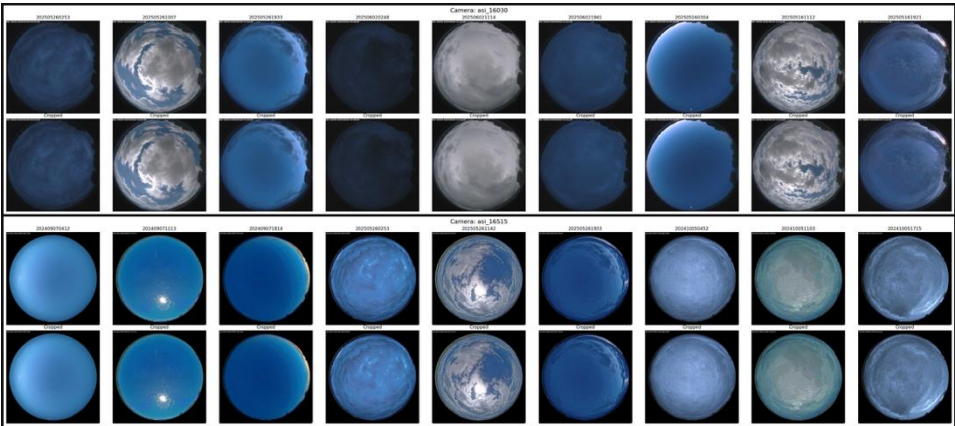


Fig. 1: Sample raw sky images from the asi_16030 (top rows) and asi_16515 (bottom rows) datasets, illustrating typical cloud cover and lighting conditions. The bottom image in each pair shows the cropped version used for processing.

5.2 Data Augmentation

To enhance generalization under real-world variability, we applied physics-plausible augmentations, including horizontal flips ($p=0.3$) to promote symmetry-invariant motion learning, Gaussian blur to simulate sensor imperfections, and ColorJitter to mimic natural sky brightness and color variations. Crucially, MixUp and CutMix were excluded, as they disrupt temporal and physical continuity key elements in structured video forecasting tasks like cloud evolution.

Original	Gaussian
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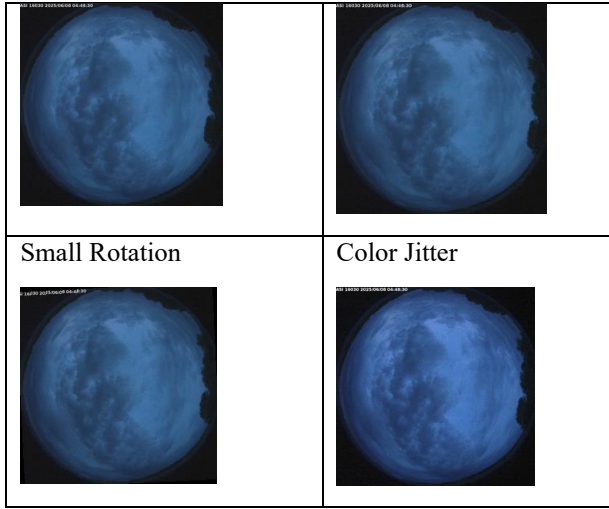


Fig. 2: Examples of physics-plausible augmentations applied to a sample sky image.

5.3 Sun Position Estimation and Masking

We used PVLib [Hol18] to compute solar azimuth and elevation per timestamp and location. A binary sun mask was generated as a white disk placed at the sun’s location over a black background. This mask was added as a fourth input channel to the RGB image, forming a (4, 128, 128) tensor. It helps the model track sun position under partial cloud cover and improves physical consistency.

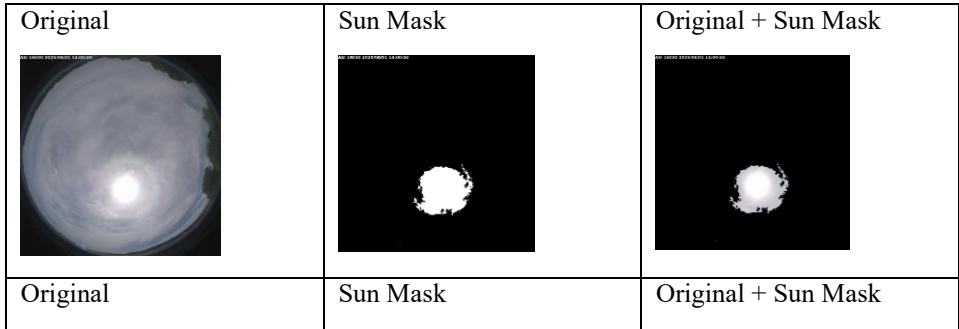




Fig. 3: Sun masking examples for clear (top) and overcast (bottom) conditions, showing original images, binary masks, and masked outputs.

5.4 Tensor Structure and Input Format

Each preprocessed sky image, including the sun mask, was resized to 128×128 , forming a (Batch, 4, 128, 128) tensor combining RGB and sun position. Input sequences consisted of 12 frames at 30s intervals (6-minute lookback), while target sequences spanned the next 30 frames (15-minute forecast).

5.5 ConvLSTM-UNet Model Implementation

The ConvLSTM-UNet baseline combines ConvLSTM layers with a U-Net encoder-decoder architecture to model spatiotemporal dynamics and reconstruct future sky frames. The encoder downsamples inputs and captures temporal patterns via ConvLSTM units, while the decoder upsamples features with skip connections to restore fine details. Implemented in PyTorch Lightning, the model predicts 30 future frames from 12 input frames, ensuring modularity, GPU acceleration, and reproducibility.

5.6 Loss Functions and Metrics

To effectively guide the ConvLSTM-UNet model's training and evaluate its performance, a composite loss function and a set of standard image quality metrics were employed:

- L1 Loss: Mean Absolute Error (MAE) was used as a primary pixel-wise loss, encouraging accurate intensity prediction across the image.
- SSIM Loss: Structural Similarity Index Measure (SSIM) was integrated into the composite loss. This perceptual metric helps to prioritize structural fidelity and improve the visually perceived quality of the predicted frames, mitigating the common issue of blurry predictions often seen with purely pixel-wise losses.

5.7 Training Setup

The ConvLSTM-UNet baseline was trained using the Adam optimizer ($\text{lr} = 1\text{e-}4$) with a batch size of 32 for 100 epochs. Training was performed on an NVIDIA A100 (80GB) via a SLURM-managed cluster. Key metrics (loss, SSIM, PSNR) and predicted video samples

were logged using Weights & Biases for detailed monitoring and visualization.

5.8 Optional GAN Refinement

An experimental **PatchGAN discriminator** was fully implemented and integrated with the ConvLSTM-UNet architecture to explore its potential in enhancing local texture sharpness and perceptual realism. We conducted ablation experiments **with and without PatchGAN**, confirming that the adversarially trained version produced noticeably sharper and more detailed predictions—a qualitative improvement, as illustrated in Section 6.

5.9 Observations: What Worked / What Didn't

The first milestone yielded key insights. PVLib-based sun masking aligned well with solar geometry. The ConvLSTM-UNet model was successfully integrated in PyTorch Lightning, and the pipeline trained reliably on the SLURM cluster. Using 30s intervals between frames helped capture cloud motion smoothly. However, limited data for rare cases (e.g., fog, sunrise, low light) may affect generalization in future stages.

6 Preliminary Results

The ConvLSTM-UNet model was trained for 100 epochs on sequences with a 15-minute prediction horizon (12 input and 30 target frames at 30s intervals). As shown in **Fig. 4**, the model without PatchGAN captures broad cloud dynamics but suffers from significant blur and lack of detail. In contrast, **Figs. 5 and 6** show predictions with PatchGAN, where visual sharpness noticeably improves, demonstrating enhanced texture and edge clarity while preserving motion dynamics. These results confirm that adversarial training reduces blurriness and improves realism.

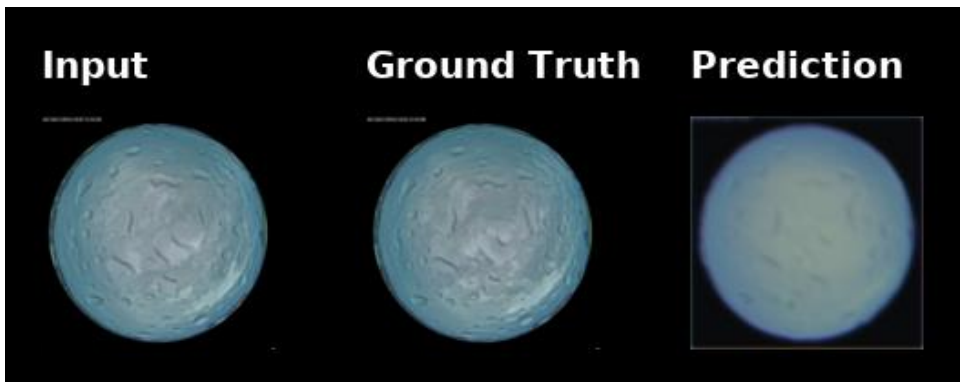


Fig. 4: Visual comparison of input, ground truth, and ConvLSTM-UNet predictions (no PatchGAN).

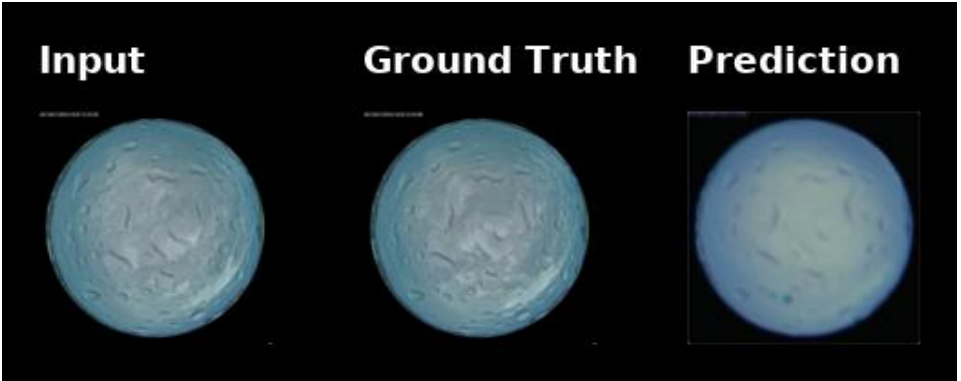


Fig. 5: Visual comparison of input, ground truth, and predicted frames.

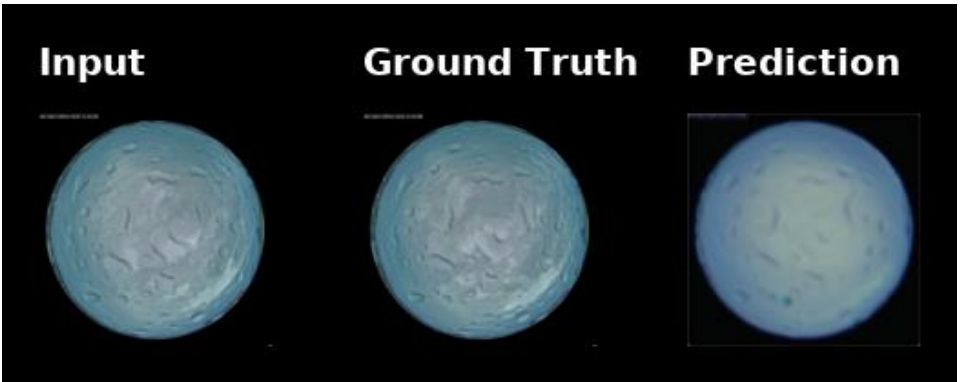


Fig. 6: Temporal evolution of predicted sky frames vs. ground truth.

Future work will expand evaluations under varied sky conditions and include per-frame error maps for detailed accuracy assessment.

7 Planned Next Steps

With the successful implementation of the ConvLSTM-UNet baseline in Milestone 1, the next steps for SkyFusion include expanding temporal forecasting, improving visual fidelity, and integrating PV power prediction. The upcoming **SkyCompressor** module will use a 3D VQ-VAE to compress sky videos into latent tokens, enabling longer forecasting while reducing computational load. The **SkyForecaster** will extend predictions with a physics-informed transformer combining GPT-style decoding and PhyCell, enhanced by PatchGAN and TemporalGAN for realism. The **SkyNowcaster** will map generated sky images into probabilistic PV estimates using CNN regressors trained

with RMSE and CRPS losses. We will benchmark SkyFusion against SkyGPT, PhyDNet, and NWP baselines using both quantitative metrics (SSIM, MAE, CRPS) and visual outputs, and assess generalization across seasons, locations, and PV surface conditions.

8 Conclusion and Impact

SkyFusion offers a modular, physics-informed pipeline for intra-hour sky and PV forecasting, extending prior work like SkyGPT with enhanced interpretability, GAN-based realism, and transformer-driven latent forecasting. This milestone delivers a reproducible ConvLSTM-UNet baseline achieving 15-minute predictions, supported by a robust data pipeline and efficient training infrastructure. While GAN integration proved valuable for enhancing visual quality, SkyFusion ultimately aims to support uncertainty-aware decision-making for grid operators, bridging visual forecasting with operational PV power needs in a scalable and explainable framework.

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